Positivity, symmetry and semi-definite optimization

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Introduction to Computational Mathematics 2022

Given a polynomial $f \in \mathbb{R}[\mathbf{x}]$ is f globally non-negative?

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Easy to check refutation (find an $x \in \mathbb{R}^n$ such that $f(x) < 0$).

Does there exist a *witness* for confirmation that is also easy to verify?

Positivity

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$f = ax^2 + bx + c \geqslant 0$ if and only if $a \geqslant 0$ and $b^2 - 4ac \leqslant 0$.
We say that f admits a sum of squares decomposition when

$$f = \sum_i f_i^2 \quad \text{for some } f_i \in \mathbb{R}[\mathbf{x}].$$

Hilbert's 17th problem

Theorem (Hilbert, 1888)

An everywhere non-negative polynomial $p \in \Sigma^2\mathbb{R}[x_1,\ldots,x_n]$ *(is a sum of squares) if and only if either*

- $n = 1$ *(univariate polynomial), or*
- $\deg p = 2$ *(quadratic polynomial), or*
- $n = 2$ *and* $\deg p = 4$ *(bivariate quartic).*

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Example

$\left(x^2+y^2+1\right)\left(x^4y^2 + x^2y^4 - 3x^2y^2 + 1\right)$ is a sum of squares!

Analytic and algebraic positivity

- $p \in \mathbb{R}[\mathbf{x}]$ is *positive* iff $p(t) \geqslant 0$ for all $t \in \mathbb{R}^n \implies$ **analytic positivity**.
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Problem

How to find such sum of squares (SOS) decomposition?

SOS decompositions

SOS decompositions

We can write f as a **quadratic function of monomials**.

Example

$$f = 4x^4 + 4x^3y - 7x^2y^2 - 2xy^3 + 10y^4$$

can be written as

$$f = \begin{bmatrix} x^2, xy, y^2 \end{bmatrix} \begin{bmatrix} 4 & 2 & -\lambda \\ 2 & -7+2\lambda & -1 \\ -\lambda & -1 & 10 \end{bmatrix} \begin{bmatrix} x^2 \\ xy \\ y^2 \end{bmatrix} = \mathbf{x}^T P(\lambda) \mathbf{x}$$

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P is so called **Gram matrix** for f.

Quadratic function of monomials

Lemma

f admits a sum of squares decomposition iff there exists a ***positive semidefinite*** *Gramm matrix for some (sub)basis* $\mathbf{x}$ *of* $\mathbb{R}[x_1, \dots x_n]$.

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If P is positive semidefinite, then $P = Q^T Q$ (e.g. by Cholesky) and

$$f = \mathbf{x}^T P \mathbf{x} = \mathbf{x}^T Q^T \cdot Q\mathbf{x} = (Q\mathbf{x})^T \cdot (Q\mathbf{x}).$$

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Example

For example for $\lambda = 6$ we have

$$P(\lambda) = \begin{bmatrix} 4 & 2 & -6 \\ 2 & 5 & -1 \\ -6 & -1 & 10 \end{bmatrix} = Q^T \cdot Q \quad \text{for } Q = \begin{bmatrix} 0 & 2 & 1 \\ 2 & 1 & -3 \end{bmatrix}$$

Therefore f admits a SOS decomposition

$$f = \left(2xy + y^2\right)^2 + \left(2x^2 + xy - 3y^2\right)^2.$$

Mathematical Programming

Linear programming:

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Semi-definite programming

- optimise linear functional
- on a polytope intersected with the cone of PSD matrices (spectrahedron)
- weak duality, non-unique solutions
- even feasibility is a hard problem!

PSD problem formulation

optimisation variables: a, b_1, b_2, c, λ.

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Example (PSD problem)

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\begin{aligned}
\text{maximise:} \quad & \lambda \\
\text{subject to:} \quad & \lambda \geqslant 0 \\
& c = 1 - \lambda \\
& b_1 + b_2 = 4 \\
& a = 2 \\
& \begin{bmatrix} c & b_2 \\ b_1 & a \end{bmatrix} \succcurlyeq 0
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tries to maximise λ as long as $(2x^2 + 4x + 1) - \lambda \geqslant 0$.

Semidefinite program

$$
\begin{aligned}
\min_{P \in \mathcal{S}^N} \quad & \langle C, P \rangle \\
\text{subject to:} \quad & \langle A_i, P \rangle = b_i, \quad i = 1, 2, \ldots, m \\
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Is $f(x) = 4x^4 + 4x^3y - 7x^2y^2 - 2xy^3 + 10y^4$ always non-negative? $\rightarrow$ sum of squares relaxation: Does there exist psd matrix P s.t.

$$f = \mathbf{x}^T P \mathbf{x} = \langle \mathbf{x}\mathbf{x}^T, P \rangle ?$$

Feasibility problem with

$$X = \begin{bmatrix} x^2 \\ xy \\ y^2 \end{bmatrix}, XX^\top = \begin{bmatrix} x^4 & x^3y & x^2y^2 \\ \cdot & x^2y^2 & xy^3 \\ \cdot & \cdot & y^4 \end{bmatrix}, A_{x^4} = \begin{bmatrix} 1 & 0 & 0 \\ \cdot & 0 & 0 \\ \cdot & \cdot & 0 \end{bmatrix}, A_{x^3y} = \begin{bmatrix} 0 & 1 & 0 \\ \cdot & 0 & 0 \\ \cdot & \cdot & 0 \end{bmatrix}, \text{ etc.}$$

$b = (b_i)$ is the vector of coefficients of f.

Problems

For n-variate polynomial of degree $2d$:

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- interior point solvers ($\mathcal{O}(mN^{3.5} + m^2N^{2.5})$)

  CSDP, DSDP, Hypatia, MOSEK, SDPA, ...

- Alternating Direction Method of Multipliers

  CDCS, COSMO, SCS, ...

- Low rank approximation of Q, (Scaled) dominantly diagonal matrices, Newton conjugate gradient, Primal-dual hybrid gradient...

Simplifying the original program

- Block-diagonalization

  (represent Q as block diagonal direct sum of psd matrices)

  1. Chordal decomposition (exploit sparsity pattern in A_is)
  2. Wedderburn(-Artin) decomposition for matrix algebras

     (group symmetry, general *-algebras, Jordan algebras, ...)

Disclaimer: This is not new, has been re-invented several times in the literature. See e.g. Gatermann, Parrilo, *Symmetry groups, semidefinite programs, and sums of squares*, 2004

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Group symmetry invariance: Example

Robinson form

$$R(x,y) = x^6 + y^6 - x^4y^2 - y^4x^2 - x^4 - y^4 + 3x^2y^2 - x^2 - y^2 + 1.$$

R is invariant under the following operations **on monomials**

$$\alpha_1 : (x,y) \mapsto (y,x)$$

$$\alpha_2 : (x,y) \mapsto (-y,x)$$

$\{\alpha_1, \alpha_2\}$ generate a set of 8 symmetries – the dihedral group D_4.

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- symmetry of monomials $\longrightarrow$ symmetry of constraints
- symmetry of monomials $\longrightarrow$ symmetry of psd matrix Q
- invariant problem = problem "remains the same" under those symmetries

Misleading quote of the day

The structure of simplifications that can be derived from a symmetry pattern does not depend on the optimization problem.

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Miracle: When $M = M_g$ are given by a linear G-action they can be **simultaneously** diagonalized to irreducibles!

Block decomposition from symmetries

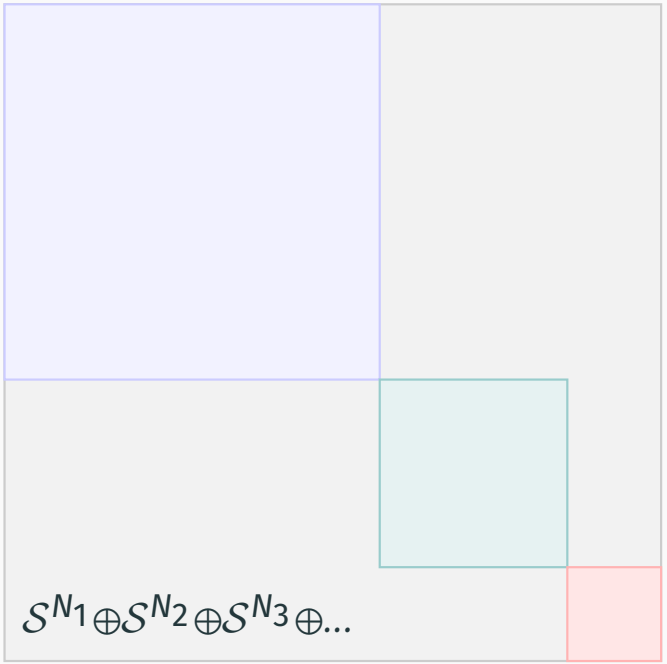
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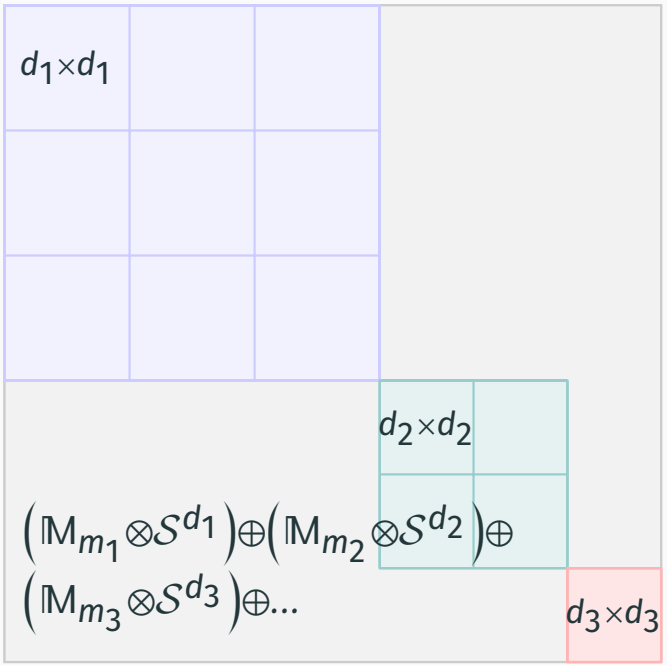
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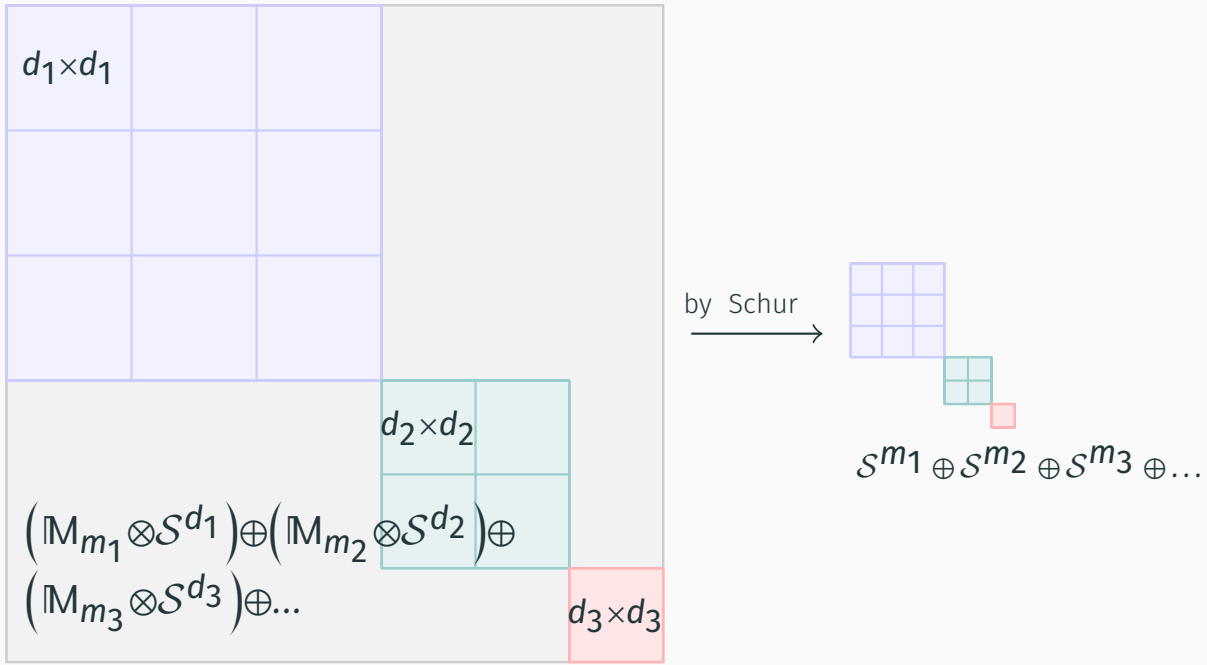
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Those isomorphims and minimal rank projections live in the group *-algebra in a basis-free form!

Example: SymbolicWedderburn.jl

```
using PermutationGroups, DynamicPolynomials
using SymbolicWedderburn
G = PermGroup([perm"(1,2)", perm"(1,2,3,4)"]) # Sym(4)
@polyvar x[1:4];  basis = monomials(x, 0:2) # 15 monomials
symmetry_adapted_basis(Rational{Int}, G, VariablePermutation(), basis,
    semisimple=true)
```

Isotypical/**semisimple** blocks when acting on `basis`:

$$B_1 = \begin{bmatrix} 1 \\ x_1+x_2+x_3+x_4 \\ x_1x_2+x_1x_3+x_1x_4+x_2x_3+x_2x_4+x_3x_4 \\ x_1^2+x_2^2+x_3^2+x_4^2 \end{bmatrix} \quad B_2 = \begin{bmatrix} x_1-x_4 \\ x_2-x_4 \\ x_3-x_4 \\ x_2^2-x_4^2 \\ x_3^2-x_4^2 \\ x_1^2-x_4^2 \\ x_1x_2-x_3x_4 \\ x_1x_3-x_2x_4 \\ x_1x_4-x_2x_3 \end{bmatrix} \quad B_3 = \begin{bmatrix} x_1x_2-x_1x_4-x_2x_3+x_3x_4 \\ x_1x_3-x_1x_4-x_2x_3+x_2x_4 \end{bmatrix}$$

We went from 15×15-psd constrain to sizes $(4\times4, 9\times9, 2\times2)$.

Example: SymbolicWedderburn.jl

```
# [ ... ]
symmetry_adapted_basis(Rational{Int}, G, VariablePermutation(), basis
  [, semisimple=false])
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Simple blocks when acting on `basis`:

$$B_1 = \begin{bmatrix} 1 \\ x_1+x_2+x_3+x_4 \\ x_1x_2+x_1x_3+x_1x_4+x_2x_3+x_2x_4+x_3x_4 \\ x_1^2+x_2^2+x_3^2+x_4^2 \end{bmatrix} \qquad B_2 = \begin{bmatrix} \frac{1}{3}(3x_1-x_2-x_3-x_4) \\ \frac{1}{3}(3x_1^2-x_2^2-x_3^2-x_4^2) \\ x_1x_2+x_1x_3+x_1x_4-x_2x_3-x_2x_4-x_3x_4 \end{bmatrix}$$

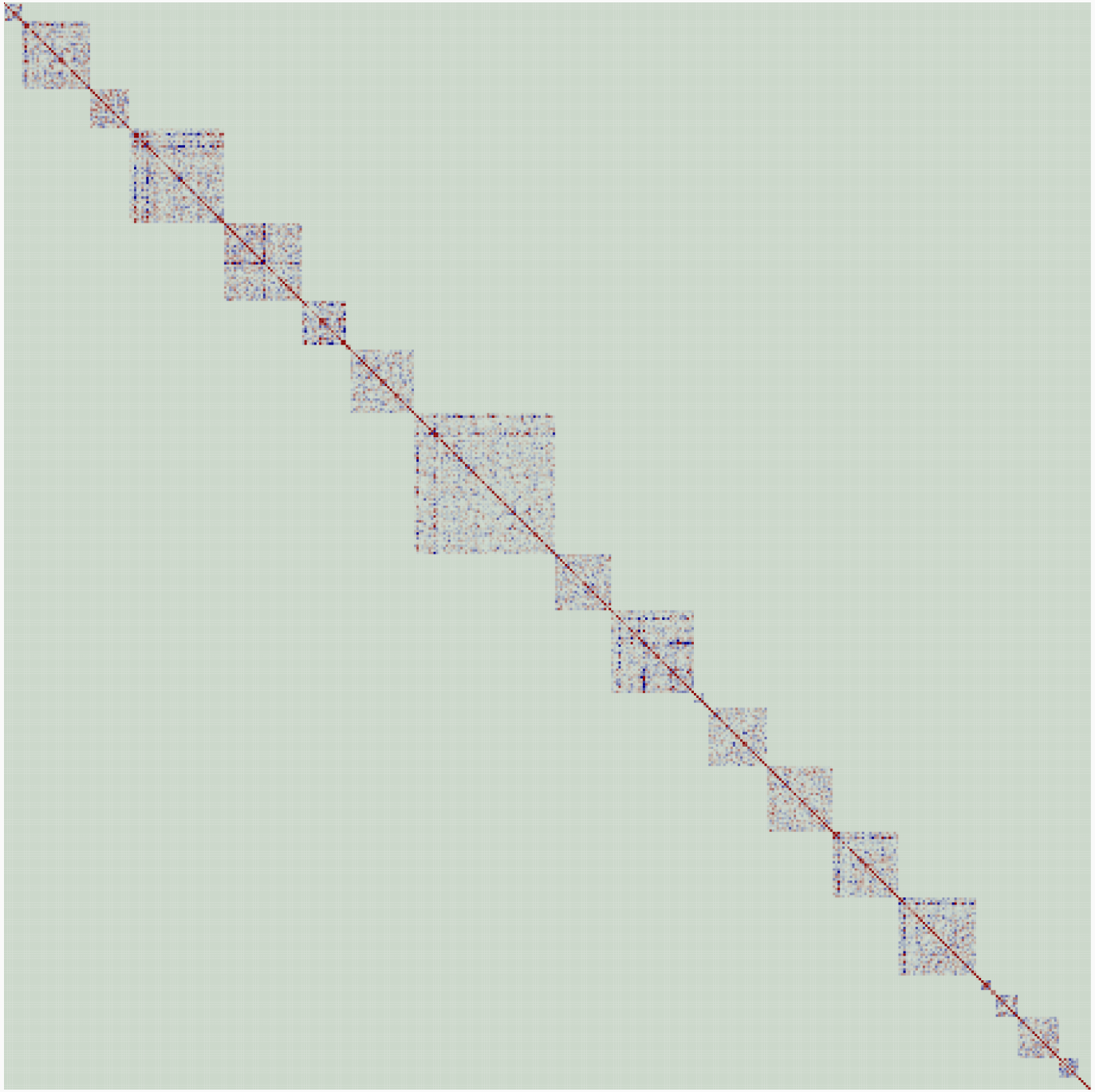
$$B_3 = \left[\frac{1}{2}(2x_1x_2-x_1x_3-x_1x_4-x_2x_3-x_2x_4+2x_3x_4)\right]$$

We went from $(4\times4, 9\times9, 2\times2)$-psd constraints to sizes $(4\times4, 3\times3, 1\times1)$.

Large scale example

- Optimization problem from geometric group theory[1]
- psd-constraint of size $4\,641 \times 4\,641$, $1.1 \cdot 10^7$ constraints
- symmetry group: $S_2 \wr S_5$ (3840 elements)
- After symmetrization:
  - 29-blocks (largest: 58×58), 13 232 variables in total
  - 7230 constraints
- Solvable in 20 minutes to $\varepsilon \sim 10^{-12}$!

[1] Kaluba, M., Nowak, P.W. & Ozawa, N. $\mathrm{Aut}(F_5)$ has property (T). *Math. Ann.* **375**, 1169–1191 (2019). https://doi.org/10.1007/s00208-019-01874-9




⟵ diagonalized psd ($\subset \mathbf{448} \times \mathbf{448}$)
Original psd constraint ($\mathbf{4\,641} \times \mathbf{4\,641}$)

```
C = ... # objective
A = ... # constraint matrices
b = ... # constraint values
basis_cnstr = ... # basis in which b is written and G acts on
basis_psd = ... # basis for the psd constraint Q

G = ... # a group, <: instance of GroupsCore.Group
action = ... # instance of SymbolicWedderburn.Action
wdec = WedderburnDecomposition(Float64, G, action, basis_cnstr, basis_psd)

m = JuMP.Model()
psds = [ JuMP.@variable(m, [1:d, 1:d] in PSDCone())
    for d in size.(direct_summands(wdec), 1)]

for iv in invariant_vectors(wdec)
    biv = dot(b, iv)
    Ai_iv = invariant_constraint(A, iv) # ~ weighted average
    As = SymbolicWedderburn.diagonalize(Ai_iv, wdec)
    JuMP.@constraint m sum(
        dot(A, P) for (A, P) in zip(As, psds) if !iszero(A)
    ) == biv
end
JuMP.@objective(m, Max, sum(dot(C, iv) for iv in invariant_vectors(wdec)))
```

Behind the scenes: `SymbolicWedderburn.jl`

- given action e.g. on monomials, induce the action to the whole basis
- compute character table of the group $\longrightarrow$ central projections
- for each central projection find minimal rank projection (symbolically, in the group *-algebra)
- evaluate the projections in the action on the basis $\longrightarrow$ matrix projections

Exploiting the symmetry: reducing constraints

Instead of using all $\langle A_i, Q \rangle = b_i$ it's enough to use

$$\left\langle \frac{1}{|G|} \sum_g g(A_i), Q \right\rangle = b_{[i]},$$

for each "invariant set" of the original constraints.

In the example: instead of looking at b_x (the coefficient at x), or b_y separately, we should be looking at $b_{\frac{x+y}{2}} = \left\langle \frac{1}{2}(A_x + A_y), Q \right\rangle$

This corresponds to finding a basis for the "invariants ring" $\mathbb{R}[x]^G$.